

Define Problem Statement

1. Problem Background

Floods cause severe damage to people, property, agriculture, and infrastructure. Traditional warning methods are often delayed or depend on limited information. An intelligent prediction system can help estimate flood risk earlier.

2. Existing System

Existing systems mainly rely on weather reports and manual analysis. They do not combine multiple environmental factors into a machine learning prediction.

3. Problem Statement

Develop an AI-powered Flood Prediction System named 'Rising Water' that predicts flood probability using environmental parameters and displays the flood risk through a simple web application.

4. Proposed Solution

The proposed system uses a trained Machine Learning model integrated with Flask. Users enter 20 flood-related parameters, and the system predicts the flood probability and categorizes the risk as Low, Medium, or High.

5. Objectives

- Predict flood probability accurately.
- Provide an easy-to-use web interface.
- Support disaster preparedness.
- Reduce decision-making time.

6. Scope

The project is intended for educational and decision-support purposes. It can be enhanced in the future by integrating real-time weather APIs and GIS maps.

7. Expected Outcome

Users receive an instant flood probability score and a corresponding risk level (Low, Medium, or High), helping them understand potential flood risk.